

Department of Computer Science and Engineering, SVNIT, Surat.
Mid-Semester Examinations, March 2025
B.Tech. III (CSE), 6th Semester
Course: Cloud Computing (CS304)

Date: 04/03/2025

Time: 10:00hrs to 11:30hrs

Max Marks: 30

Instructions

- Please start the answer to each question on new page ONLY of your answer sheets
- Please write your correct exam number without fail on the answer sheets as well as on the question paper
- Please be brief and to-the-point in the attempting the answers to the theory questions. Unnecessary and unjustified elaboration will not fetch more marks. The hallmark of an answer to a short question indeed is its brevity

Q.1 Answer the following:

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|---|--|----|
| 1 | Explain following terms in brief: VMM, CAP theorem, SLA, Container Engine | 04 |
| 2 | List out the resources which can be virtualized. Explain any two resource virtualizations in detail. | 04 |
| 3 | Compare and contrast cloud deployment models with examples. | 06 |
| 4 | Which cloud service model is preferred when security is important for varying requirements? Write its characteristics, advantages and limitations. | 04 |

Q.2 Answer the following:

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|---|---|----|
| 1 | A cloud provider is using QEMU/KVM to host VMs for a high-performance database application. The application experiences high latency during I/O operations. How does QEMU/KVM optimize I/O virtualization to reduce latency? | 02 |
| 2 | A cloud provider is running multiple VMs on a hypervisor that dynamically adjusts memory allocation between VMs. One VM, running a memory-intensive application, starts crashing frequently, even though the host system reports having free memory. The guest OS shows high memory usage, while the hypervisor logs indicate that memory is being reclaimed from the VM despite its high demand. What memory reclamation policy it is using currently. What could be a better memory reclamation policy that can help in this situation and why? | 02 |
| 3 | A developer is testing a new OS on a VM using a hypervisor that does not support hardware-assisted virtualization. The OS crashes when it tries to execute privileged instructions, even though the same code runs fine on physical hardware. Why does this happen? How does the hypervisor handle privileged instructions in this scenario? What techniques can be used to resolve the issue without modifying the OS source code or hardware? | 04 |
| 4 | How does the execution of a guest application differ between QEMU/KVM and full virtualization? Discuss the differences in privilege levels, handling of privileged instructions, execution flow, and performance impact. How do these approaches affect the efficiency of guest applications? | 04 |
